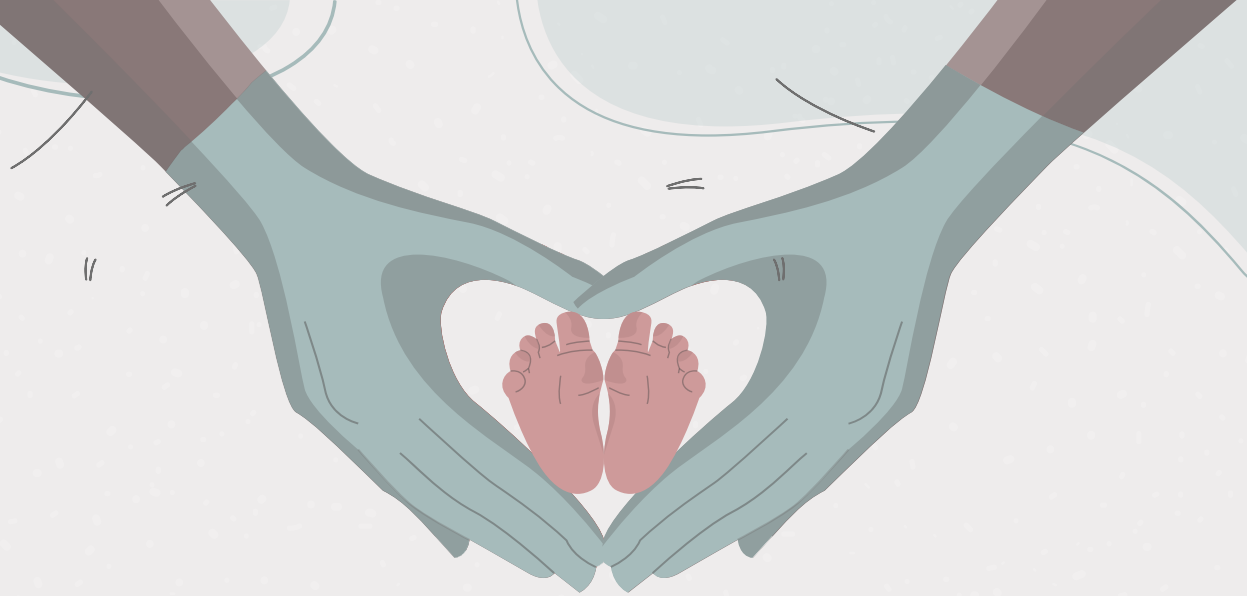
GastroWave

Neonatal Belt

Treating GERD in Premature Infants

Team Bio Wizards



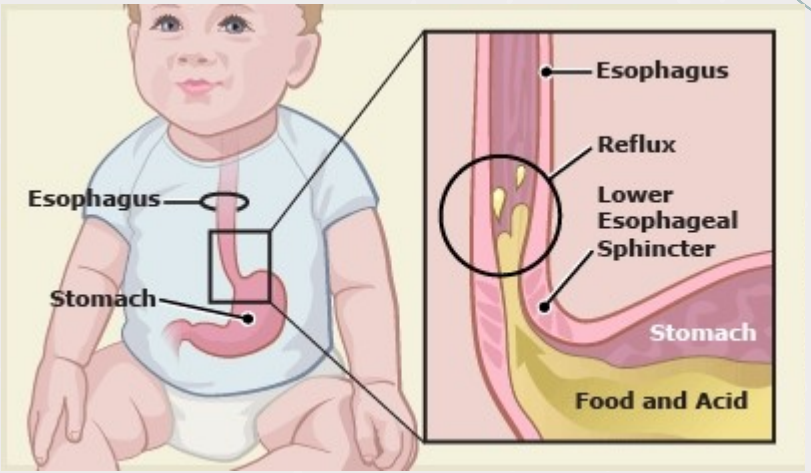


01 Background

Gastroesophageal Reflux Disease (GERD)

- Immature lower esophageal sphincter (LES) and underdeveloped GI motility → **stomach contents flow back into the esophagus**
- Frequent reflux episodes in NICU → feeding intolerance, poor weight gain, and increased risk of aspiration → **hindered weight gain** and **increased length of stay (LOS)**

GERD is a common and significant clinical challenge in the care of premature infants.





1 in 10 [1]

Preterm infants diagnosed with GERD

2-3 episodes [2]

Of reflux per hour

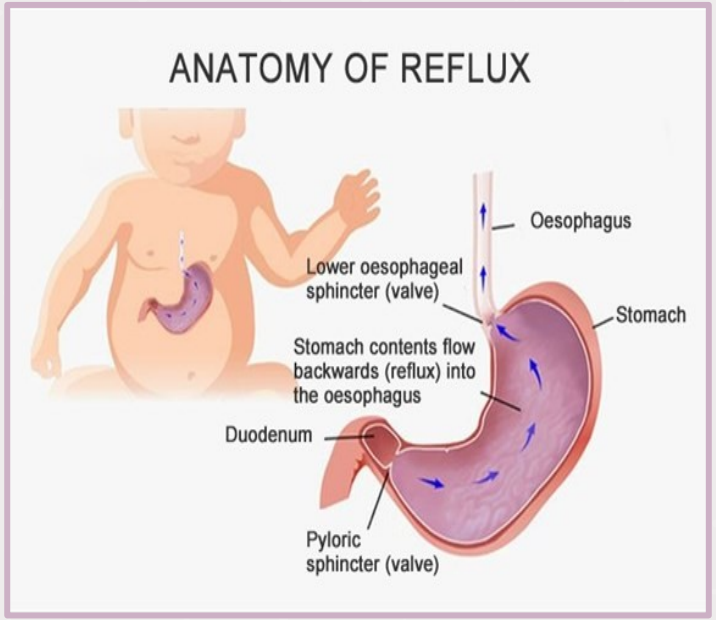
$70,000 [3]

Added per discharge of baby diagnosed with GERD

Anatomy

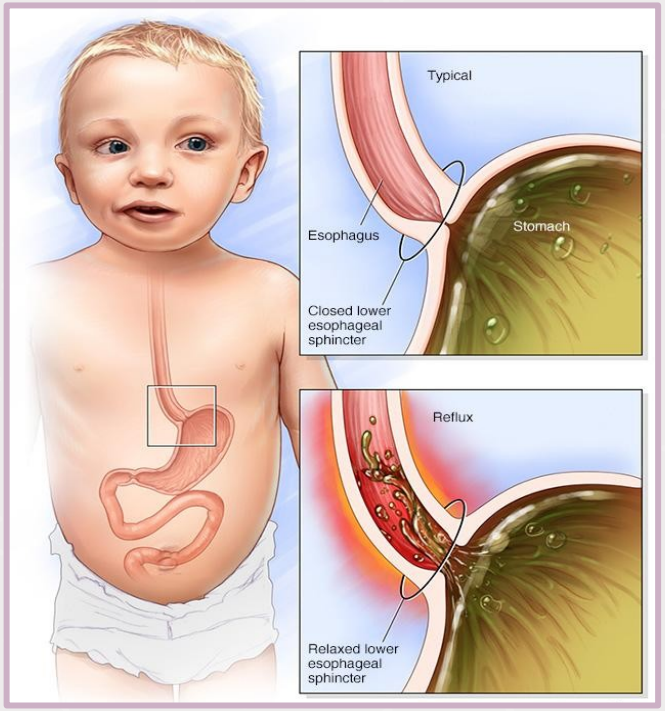
- **Lower Esophageal Sphincter (LES):** Prevents backflow to the esophagus
- **Diaphragm:** Maintain Pressure Differences
- **Esophagus:** Tube that connects mouth to the rest of the digestive system
- **Peristalsis:** Contractions to move food downward
- **Weak LES + high stomach pressure:** reflux episodes

[4]



Physiology[5]

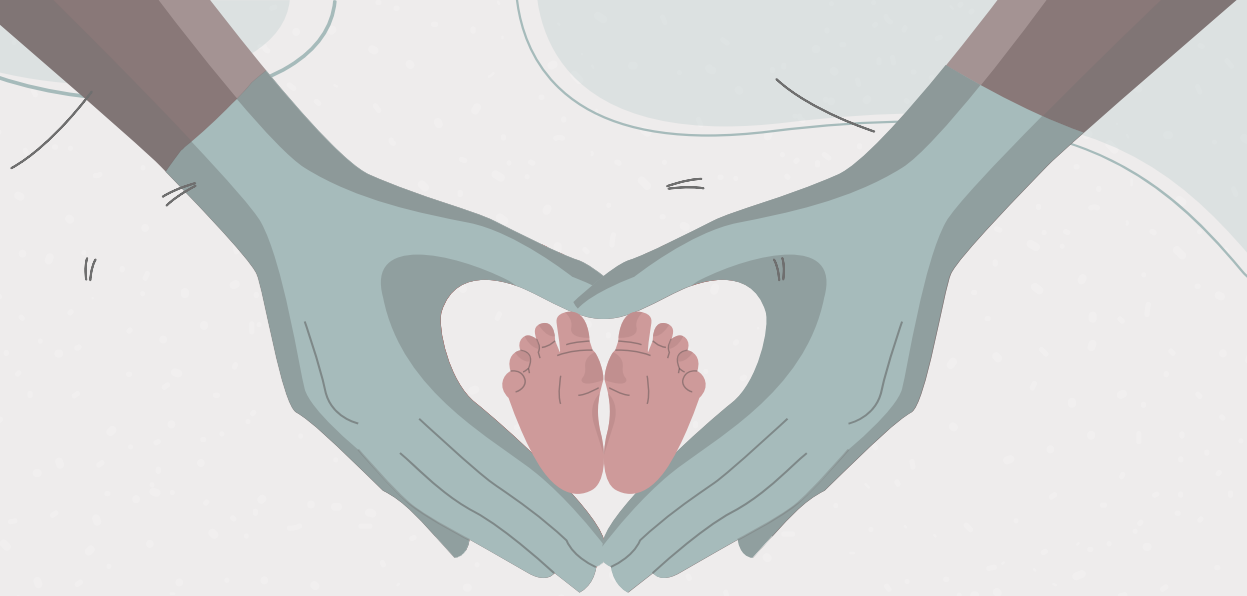
- Premature infants have **incomplete digestive systems**
- Milk stays in the stomach longer, which **increases stomach pressure**
- **Weaker peristalsis and flat position** pushes contents back to mouth through the esophagus



Problem Statement

“To develop a non-invasive gastric stimulator for premature infants that reduces GERD while addressing the invasiveness, inconsistency, and nurse-dependence of current treatments.”





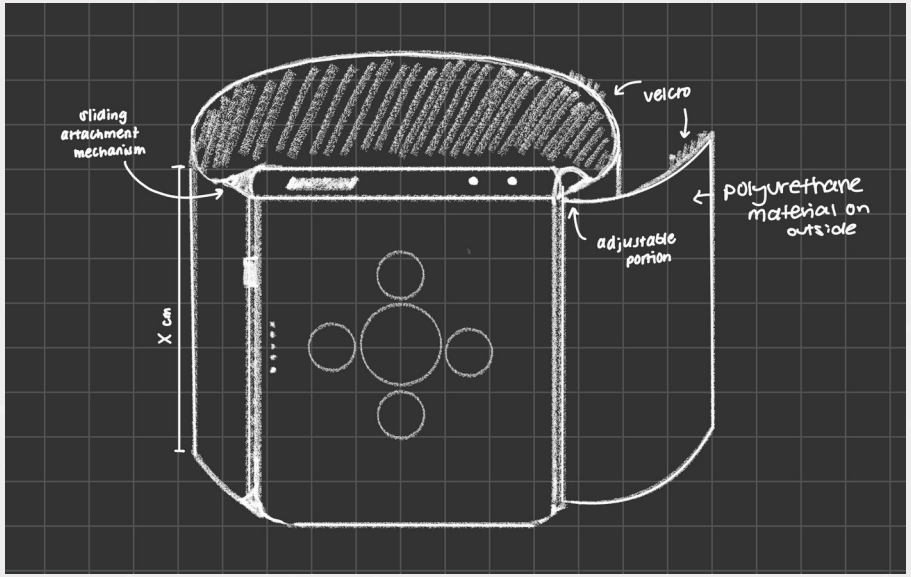
02 Design

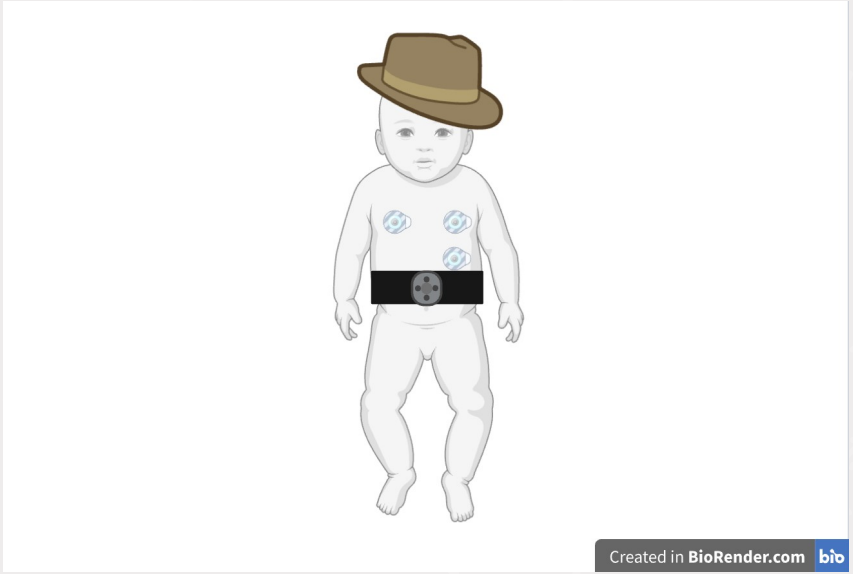
Design Constraints

- Temperature probes and cardiac leads on abdomen
  - Gentle pressure should not interfere with temperature or heart activity readings
- Nurse workflow is tight, avoid additional lengthy procedures
  - Device should be able to be worn for extended periods of time and activate autonomously

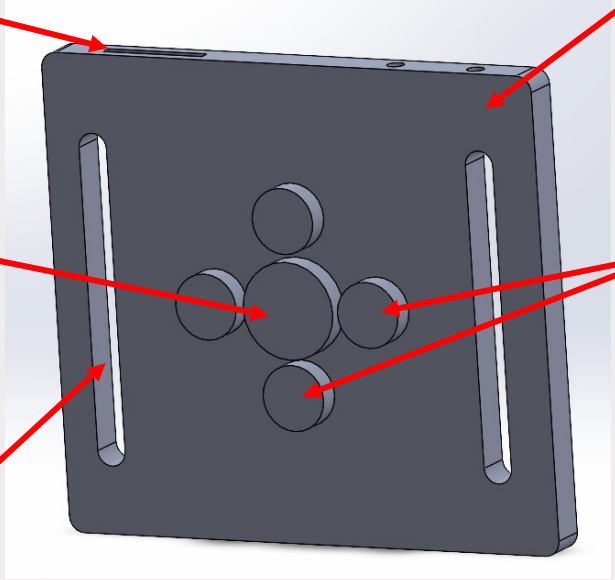


Product Sketch





CAD Model & Features



1 Charging Port

Rechargeable lithium ion battery for multiple uses

2 Pressure Sensor

Ensures force does not exceed 0.5 - 1.0 N, sensors are needed, activates if abdomen is distended

3 Fabric Belt Slots

Insertion site for soft polyurethane belt for anti-chafing security

4 Thermo-Polyurethane Body

Can be sterilized with EtO (ethylene oxide) after each use

5 Soft Pneumatic Cells

To generate clockwise movement, soft cells are used

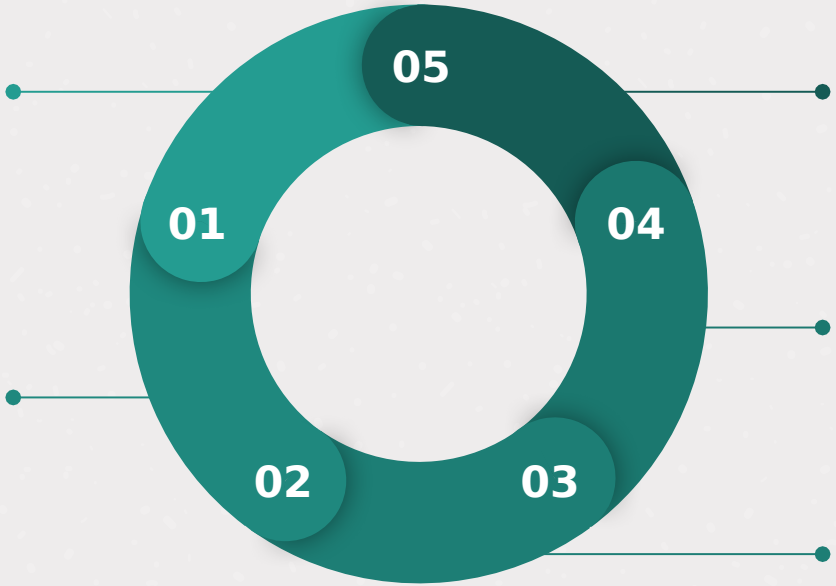
6 Pressure Relief System

To avoid excess pressure buildup, pressure relieving valves are built

How GastroWave Works

Step 1 - Turn on the device
Turn on the power to activate controller

Step 2 - Pump activates
Internal micro-pump is activated

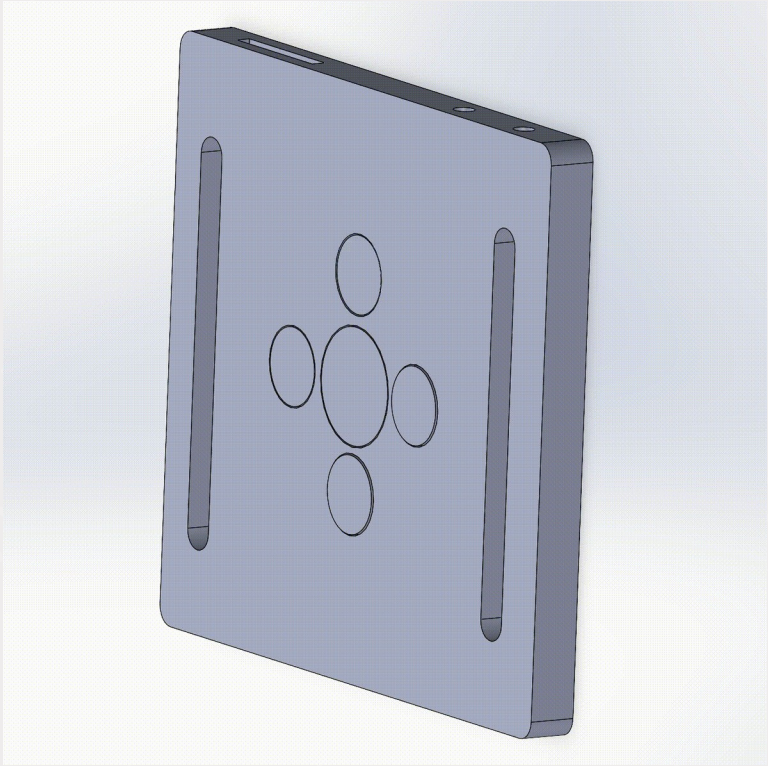


Step 5 - Turn off the device
Device is turned off

Step 4 - Repeat
Device can be kept turned on until satisfactory results are achieved

Step 3 - Pneumatic cell movement
Cell 1 pops up, rotates, goes back in. Repeated for each cell to complete one cycle.

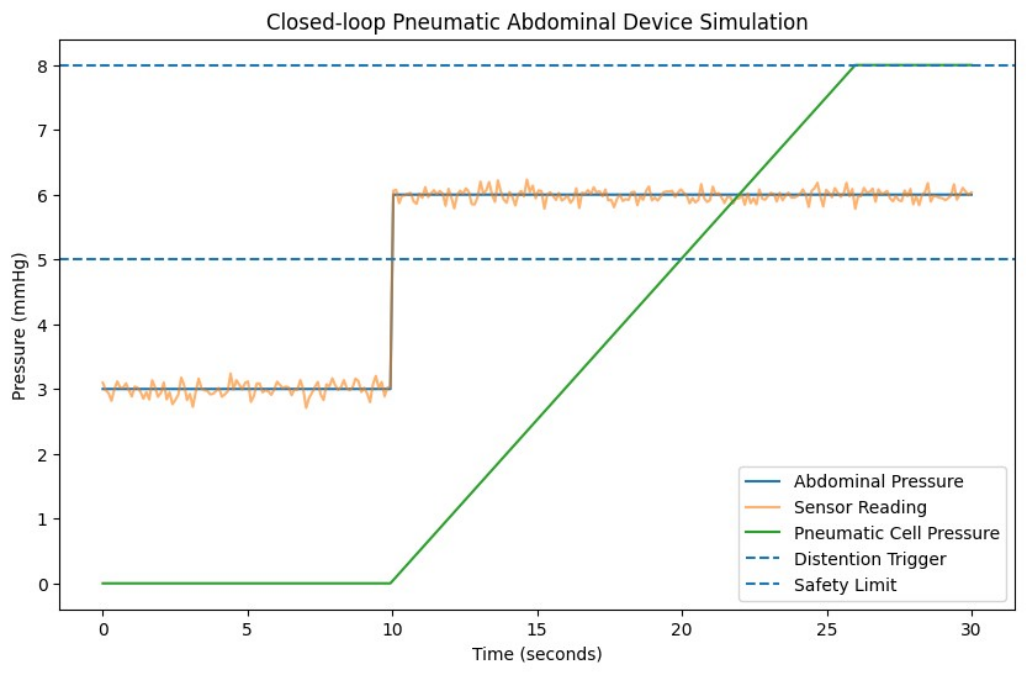
CAD Animation



Clockwise motion to simulate peristalsis and emulate physical massage.

Note: Rotation of pneumatic cells is not shown due to rendering limitations

Python Simulation for Closed-Loop



What this simulation shows:

- Baseline pressure (~3 mmHg)
- Distention event (~6 mmHg)
- Pump activation when threshold exceeded
- Pneumatic pressure buildup
- Automatic shutdown at safety limit
- Validates closed-loop safety control.

The simulation demonstrates that GastroWave can detect abdominal distention, activate stimulation, and automatically stop before reaching unsafe pressure levels.

Inclusivity

Sizing & Adjustability



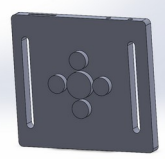
Small: 7.5-9 cm² contact area

- Target Weight: **(500 - 800 g infants)**
- Abdominal circumference: **16–20 cm**
- Designed for extremely premature infants (<28 weeks gestation)
- Reduced pneumatic pressure to accommodate fragile abdominal tissue



Medium: 9-10.5 cm² contact area

- Target Weight: **(800 g - 1.5 kg infants)**
- Abdominal circumference: **20–24 cm**
- Standard NICU premature infant range
- Balanced stimulation pressure to promote gastric motility



Large: 10.5-12 cm² contact area

- Target Weight: **(1.5 - 2.5 kg infants)**
- Abdominal circumference: **24–28 cm**
- Late-preterm or recovering NICU infants
- Maintains consistent stimulation coverage across larger abdomen

Minimum Viable Product Criteria

Area of Device

- 7.5 cm^2 - 12 cm^2

No. of Cycles Before Failure

- 30,000 cycles before failure

Battery Life

- Rechargeable
- At least 10 hours of function before recharging

Cell Time

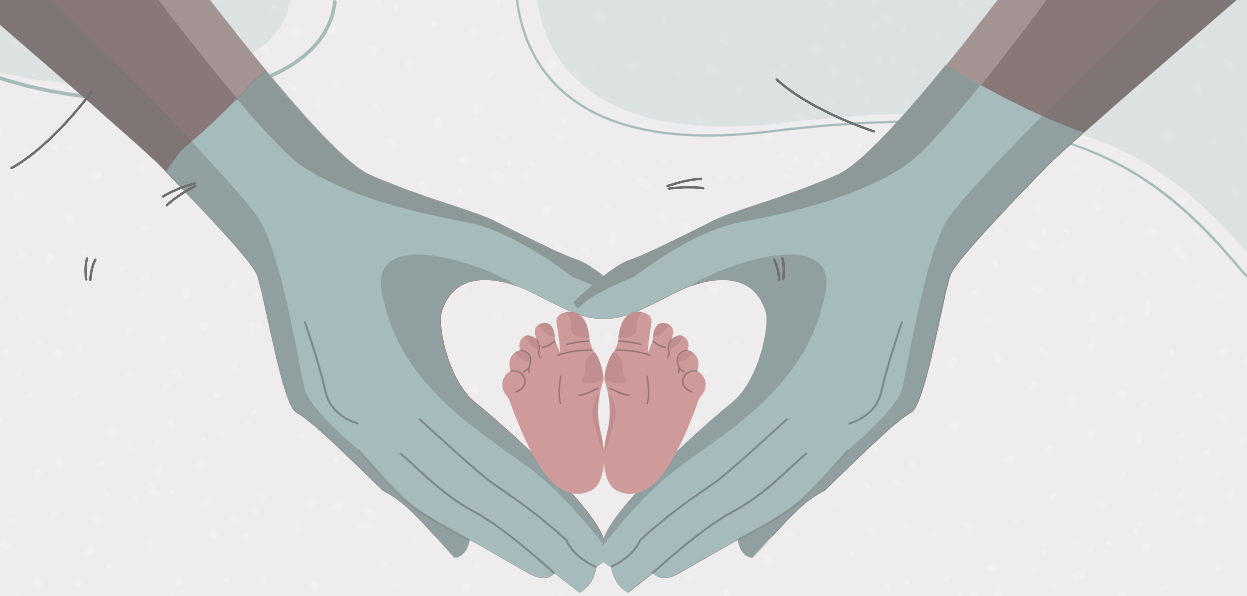
- Inflation: <8 sec
- Deflation: <4 sec

Weight

- Maximum ~ 50 g

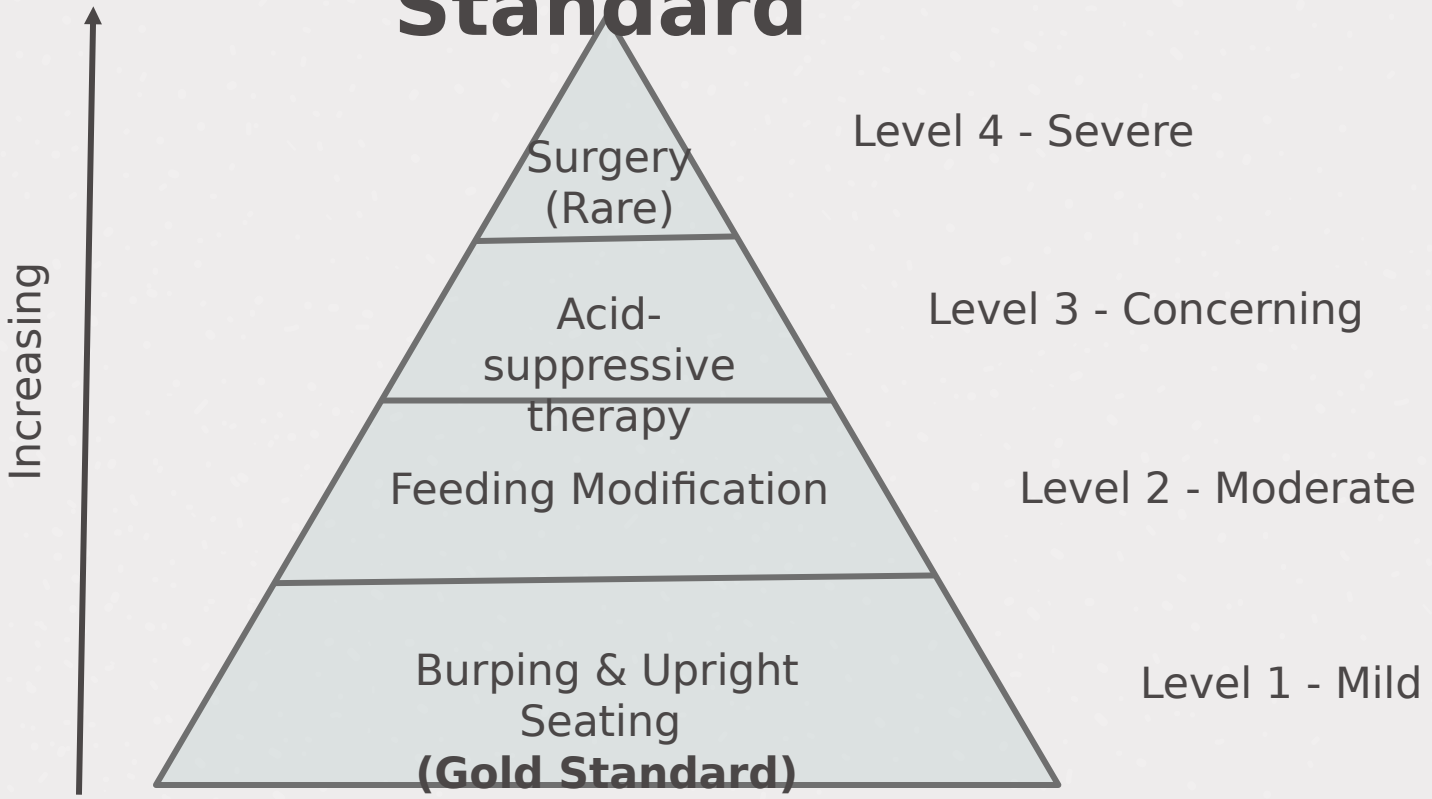
Others

- Heat Resistant
- Biocompatible
- Sterilizable (EtO)
- Leak Resistant



03 Competition

Current Treatment/ Gold Standard



Pugh Matrix with Current Solutions

Criteria	Baseline: Upright Seating and Burping	Acid-Suppressive Therapy	Diet Modification	GastroWave
Safety	0	-1	0	1
Effectiveness	0	1	0	1
Non-Invasiveness	0	1	-1	1
Cost	0	-1	-1	1
No Infection/ Side-Effects	0	-1	-1	1
Ease of use	0	0	0	1

Impact on Premature Infant Care

01 Weight Gain

Improved gastric emptying helps premature infants absorb nutrients and gain weight more effectively. [20]

02 Improved Sleep

Fewer reflux episodes reduces discomfort and allows infants to maintain longer, stable sleep cycles. [18]

03 Reduced Medication Dependence

Non-pharmacological solutions reduces reliance on acid- suppressive medications and their potential side effects. [20]

04 Reduced Nurse Workload

Standardized reflux management reduces the need for constant repositioning and manual intervention. [19]

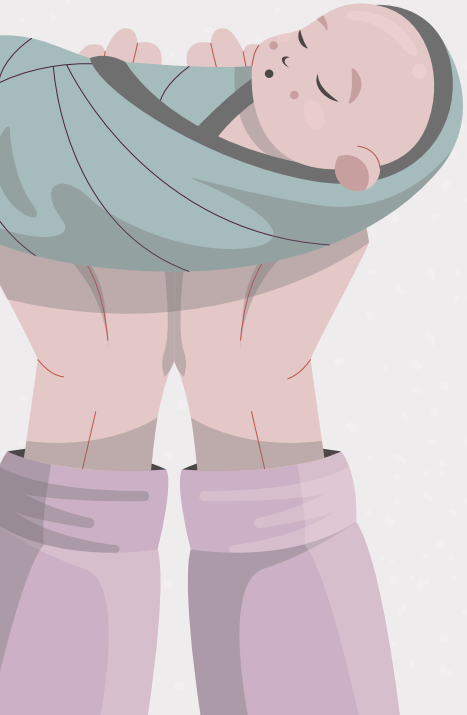
05 Improved Feeding Tolerance

Gentle abdominal stimulation promotes gastric motility, helping infants tolerate feeding with fewer regurgitation events. [21]

06 Lower Chance of Aspiration

Reducing reflux decreases the risk of milk entering the airway, preventing aspiration and respiratory complications. [20]

Feasibility





Skin Electrode Abdomen Stimulator [7]

Uses electrode, pulse generator and control unit to stimulate abdominal movement



FDA-approved Abdominal TENS Stimulation Belt [27]

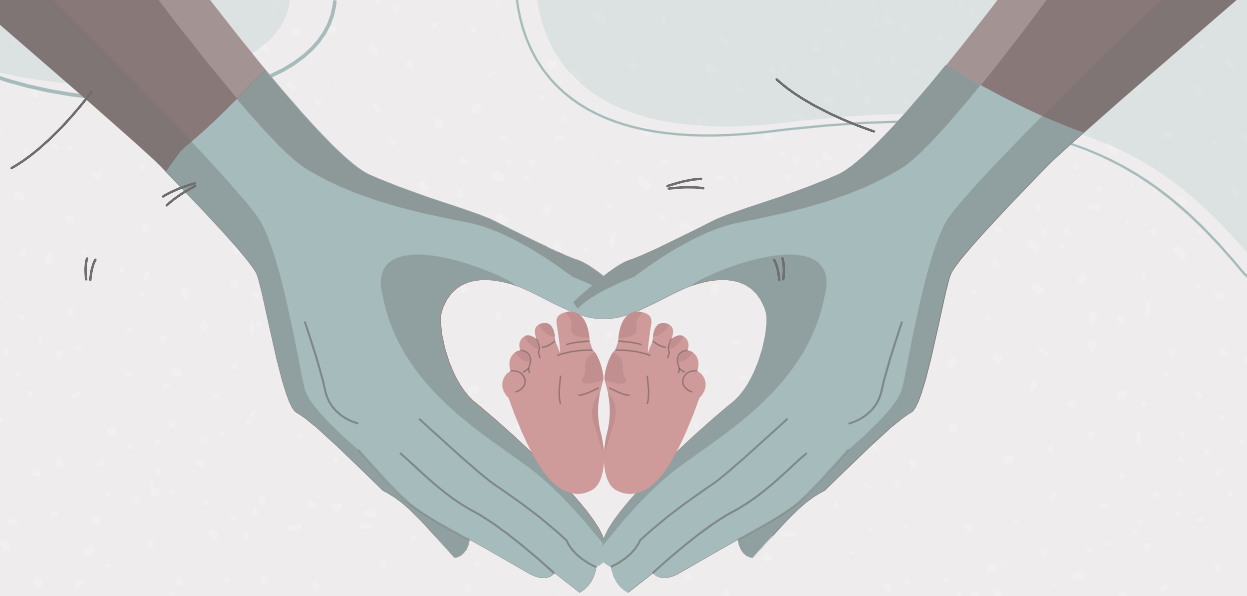
Non-invasive electrical stimulation activates abdominal muscles to promote gastrointestinal motility



Harness to Maintain Position

Harnesses and straps that help maintain upright positions

Our innovation integrates gastric stimulation with pressure feedback control.



04 Regulatory

Path to FDA Clearance

Classification: Class II

High-risk patient population, must provide evidence that no harm is done and prove efficacy of safety measures [23]

- Confirm classification by submitting 513(g) for FDA feedback [26]



Pathway: De Novo

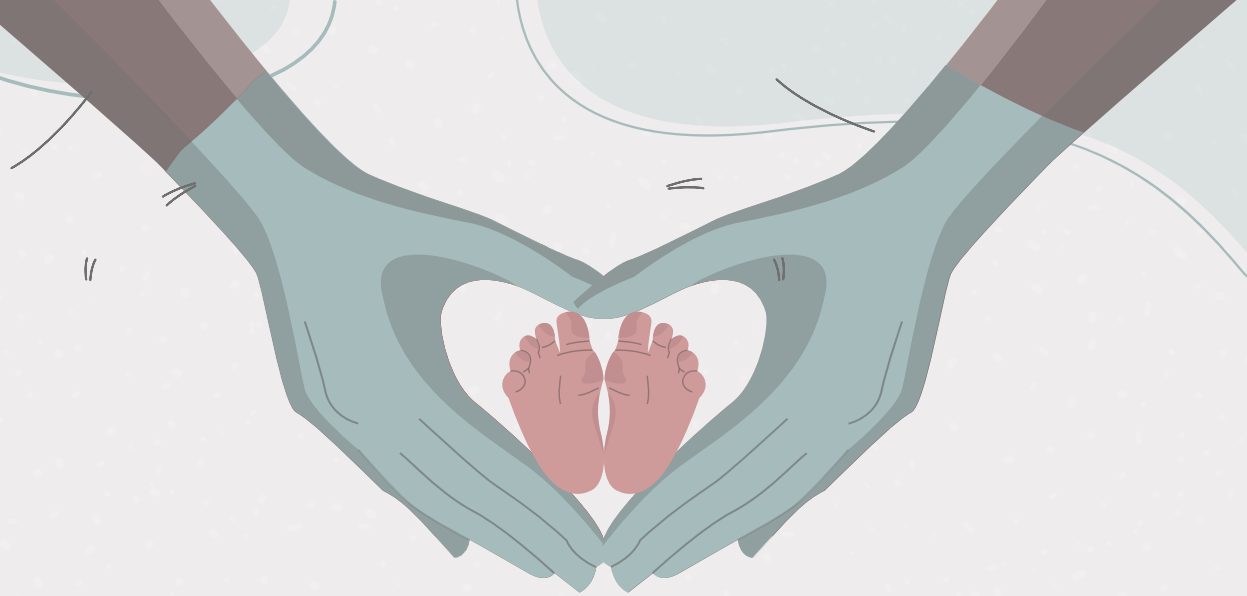
No predicate device similar enough for 510(k) clearance, but not high-risk enough to need pre-market approval

Submission Considerations:

- NSE Predicate: TENS Muscle Stimulation Belt [27]
- Clinical trials to show safety and efficacy
- Prove minimum viable product criteria are met

ISO Standards Guiding GastroWave Development

Standard	Why It Applies	Engineering Validation Benchmarks
ISO 10993	Biocompatibility for neonatal skin contact	Skin contact materials tested per biocompatibility standards
ISO 14971	Risk management for abdominal pressure & monitoring interference	Max abdominal pressure ≤ 1 N
ISO 11607	Sterile packaging & ethylene oxide sterilization	Device sterilization via ethylene oxide
ISO 13485	Quality control during manufacturing	Manufacturing traceability & quality control implemented
IEC 60601-1	Electrical safety and electromagnetic compatibility with NICU monitoring equipment	Electrical safety validated



05 Market Analysis

Target Market & Audience

Target Market

Premature Infants < 28 Weeks

Target Audience

Primary Customer
Hospitals with NICU units

Initial deployment
Urban hospitals in the U.S.

Expansion
Rural U.S and Global NICUs in low- and middle-income countries



TAM, SAM, SOM

$19.4M

Total Addressable Market (TAM)

Total market demand if all premature infants with GERD used the device

- 10% of total births are preterm = 370,000 premature infants/year
- 50% chance of GERD = 194,000 infants
- $100 per device
- TAM = $19.4 million/year

$9.7M

Serviceable Available Market (SAM)

Market demand focusing only on premature infants with GERD in NICU

- 10% of total births are preterm = 370,000 premature infants/year
- 51% preterm infants are admitted to NICU = 194,000 infants
- 50% chance of GERD = 97,000 infants
- $100 per device
- SAM = $9.7 million/year

$630K

Serviceable Obtainable Market (SOM)

Realistic market demand focusing only on premature infants with GERD in NICU in Urban America in only one city (~5%)

- 50% chance of GERD = 97,000 infants
- 5% adoption rate in first year in urban America (5% of 65%) = 6,300 infants
- $100 per device
- SOM = $630,000/year

TAM, SAM, SOM - Summary

TAM
All premature infants with GERD
$19.4M

SAM
NICU treated infants
$9.7M

SOM
First city rollout
$630K

Bill of Materials

Component	Price	Notes
Air Pump and Vacuum DC Motor [15]	$7.95 per pump/motor	Could use multiple pads with one pump
TPU Bladder Film	$6.16 per 25 cm^2 sheet	Sheet would be sectioned into 4 pads connected to a single motor
Medical Grade PVC Tubing	~$1.99 per foot	Bulk orders could further reduce prices
TP4056 (Li-Ion Charger)	~$7.49 per pack of 5	

Bill of Materials (Cont.)

Component	Price	Notes
Polyurethane Laminated Fabric [13]	~$7.45 per yard in bulk	Need less than 0.025 yards^2 for one unit
Li-Ion Coin Cell Battery	$2.92	Sheet would be sectioned into 4 pads connected to a single motor
Hook-and-Loop (Velcro) Fasteners [14]	$9.47 for 4 ft	Bulk rolls can reduce pricing to <$0.5 per foot

Cost to Assemble + Distribute

Component	Price	Notes
Sewing + Assembly Labor	$5–$8 per garment in large-scale factories	Labor typically represents 20–30% of fabric production cost [28]
Estimated Total Manufacturing Price	**~$51.43** total	Includes 12% overhead to account for additional costs
Regulatory and Distribution Costs	+50% (Regulatory), +30% (Distributor Markup), **~$92.57 PPU**	R&D/Clinical Trials/FDA approval can cost millions over years

IP/Patent Strategy

Provisional Patent

- Cheaper to file
- Gives 12 months for logistics
- Idea Protection

File Utility Patent

- Utility of soft pneumatic cell technology to tackle GERD



Find Sources to Fund and File Patent

- Patents can be expensive, participate in more competitions, funding rounds or join incubator/accelerator programs
- Find IP lawyer/firms

Exit Strategy

Get Acquired

Benefits -

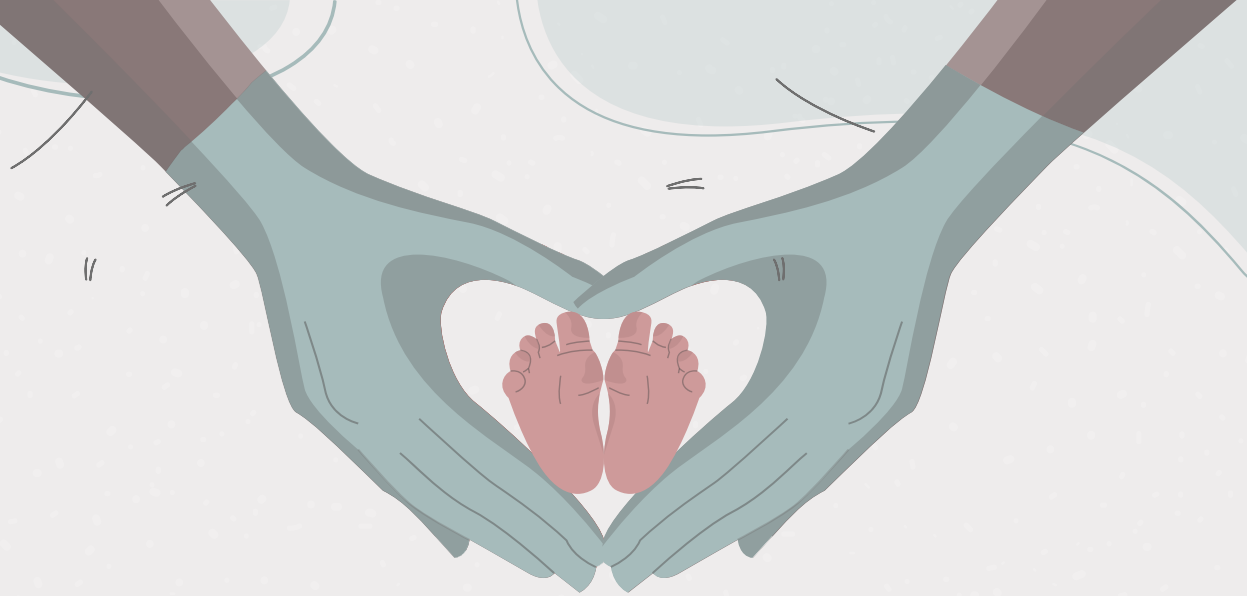
1. Lesser Capital Required than IPO
2. Access to Hospital Network, R&D, Manufacturing and Supply Chains

Medium Companies

- Masimo
- Natus Medical
- Atom Medical
- Inspiration Health Group

Large Companies

- Medtronic
- Philips
- Drager
- G.E Healthcare



06 Reference & Appendix

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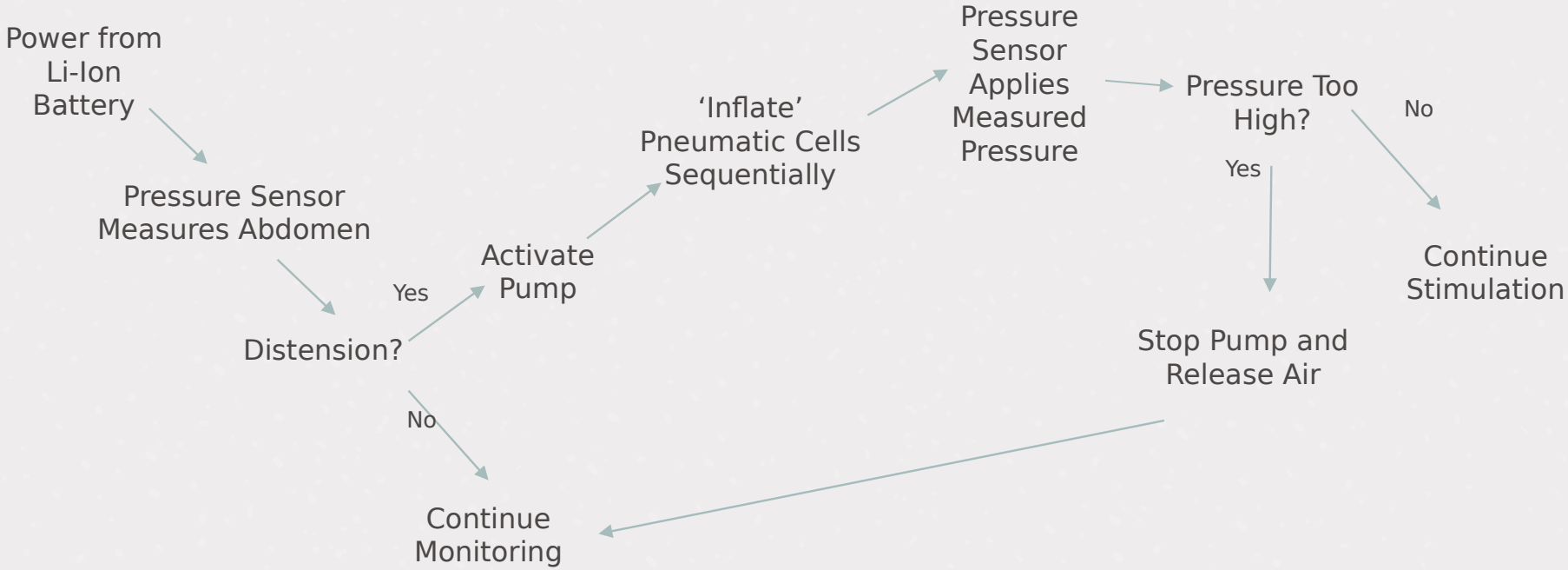
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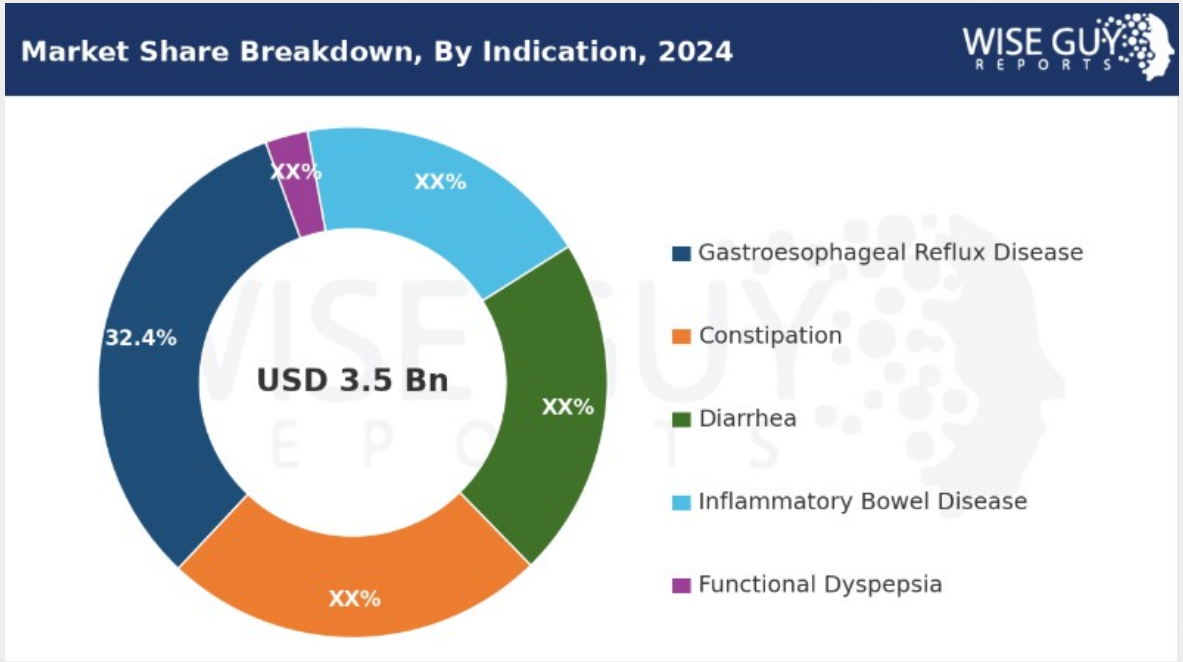
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Appendix - Workflow of the GastroWave

Distension Categories:
Baseline: 2–4 mmHg
Distention trigger: +1–2 mmHg change
Maximum allowed stimulation: 6 mmHg
Emergency cutoff: 8 mmHg
Source: [22]



Appendix - Gastrowave Market Penetration



Thank You!
Questions?

